

Positive Definiteness of the Semidiscrete Moment-Measure Newton Matrix

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Status

This is a partial result for Bonnet–Rubinstein, *Quantitative Stability and Numerical Resolution of the Moment Measure Problem*, arXiv:2604.09914 [1]. The source paper leaves two questions for future work in Section 4.3: invertibility of the regularized Newton matrix used in Algorithm 4.1, and convergence of the damped Newton method. The result below proves the invertibility part at every full-cell iterate $\Phi \in U$. It does not prove global convergence, nor does it address how far the iterates remain from the boundary of U .

Source Passage

The source defines the regularized matrix on page 22:

4.3 Damped Newton method

In view of the above computations, one can attempt to compute a minimizer of E_ν using a damped Newton method, that we describe as Algorithm 4.1. Observe that this method is largely inspired by a similar approach used in the setting of the optimal transform problem, see e.g. [18].

Algorithm 4.1 Damped Newton method for computing ψ_ν

Require: a tolerance $\varepsilon > 0$, an initialization $\Phi^{(0)} \in U$.

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for  $k \in \mathbb{N}$  do
  if  $\|\nabla E_\nu(\Phi^{(k)})\| \leq \varepsilon \|(\nu(\{y_i\}))_{1 \leq i \leq N}\|$  then
    return  $(\Phi^{(k)})^* + \log \int_{\mathbb{R}^d} e^{-(\Phi^{(k)})^*(x)} dx$ 
  end if
  find  $d^{(k)} \in \mathbb{R}^N$  s.t.  $\nabla^2 E_\nu(\Phi^{(k)})d^{(k)} = -\nabla E_\nu(\Phi^{(k)})$ 
   $\tau^{(k)} \leftarrow \max\{2^{-i} \mid i \in \mathbb{N}, \Phi^{(k)} + 2^{-i}d^{(k)} \in U\}$ 
   $\Phi^{(k+1)} \leftarrow \Phi^{(k)} + \tau^{(k)}d^{(k)}$ 
end for

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Algorithm 4.1 is a damped Newton method, since the step size is sometimes reduced to ensure that all iterates belong to the set U . This is motivated by the fact that the formulas (4.7) and (4.8) for the elements of $\nabla^2 E_\nu(\Phi)$ are only valid when $\Phi \in U$. Remember that Proposition 4.5 guarantees that the minimizer of E_ν belongs to U .

The matrix $\nabla^2 E_\nu(\Phi^{(k)})$ appearing in Algorithm 4.1 is not invertible, since E_ν is invariant under addition of a vector of the form $(a + \langle v, y_i \rangle)_{1 \leq i \leq N}$, for $a \in \mathbb{R}$ and $v \in \mathbb{R}^d$. In practice, we rather solve the linear system $M_\nu(\Phi^{(k)})d^{(k)} = -\nabla E_\nu(\Phi^{(k)})$ with $M_\nu(\Phi^{(k)}) := \nabla^2 E_\nu(\Phi^{(k)}) + uu^\top + \sum_{i=1}^d v_i v_i^\top$, where u denotes the vector of size N whose all elements are one, and v_i denotes the vector $((y_j)_i)_{1 \leq j \leq N}$. According to (4.7) and (4.8), $\nabla^2 E_\nu(\Phi^{(k)})$ is the sum of a sparse matrix and a matrix of rank one, and thus $M_\nu(\Phi^{(k)})$ is the sum of a sparse matrix and a matrix of rank $d + 2$. Elements of $\nabla E_\nu(\Phi^{(k)})$ and $\nabla^2 E_\nu(\Phi^{(k)})$ (and thus $M_\nu(\Phi^{(k)})$) can be computed using (4.6), (4.7), (4.8), and exact integration formulas for the integrals over Laguerre cells.

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The explicit future-work sentence appears at the top of page 23:

The question of the invertibility of the matrix $M_\nu(\Phi^{(k)})$, and of the convergence of the method described in Algorithm 4.1, is left for future work. The analysis of similar Newton methods in the setting of optimal transport [18] suggests that it may be useful to modify the damping criterion to impose that the iterates are sufficiently far from the boundary of U ; such considerations are also left for future work. In practice, on our experiments, we do not observe any issues with the damping criterion, nor with solving the linear system $M_\nu(\Phi^{(k)})d^{(k)} = -\nabla E_\nu(\Phi^{(k)})$ using a sparse linear solver.

5 Numerical experiments

Transcription of the Target

In Section 4.3, after defining

$$M_\nu(\Phi) = \nabla^2 E_\nu(\Phi) + uu^\top + \sum_{\alpha=1}^d v_\alpha v_\alpha^\top,$$

where $u = (1, \dots, 1)$ and $v_\alpha = ((y_j)_\alpha)_{1 \leq j \leq N}$, Bonnet–Rubinstein write:

The question of the invertibility of the matrix $M_\nu(\Phi^{(k)})$, and of the convergence of the method described in Algorithm 4.1, is left for future work.

Definitions

Let $\nu = \sum_{i=1}^N p_i \delta_{y_i}$ be a finitely supported probability measure on $\mathbb{R}^d$, with all $p_i > 0$, centered, and not supported on a hyperplane. Thus the vectors

$$u = (1, \dots, 1), \quad v_\alpha = ((y_i)_\alpha)_{i=1}^N, \quad 1 \leq \alpha \leq d,$$

are linearly independent in $\mathbb{R}^N$. For $\Phi \in \mathbb{R}^N$, put

$$\Phi^*(x) = \max_{1 \leq i \leq N} (\langle x, y_i \rangle - \Phi_i)$$

and

$$E_\nu(\Phi) = \sum_i p_i \Phi_i - \log \int_{\mathbb{R}^d} e^{-\Phi^*(x)} dx.$$

Let U be the open set of Φ for which every Laguerre cell

$$\text{Lag}_i(\Phi) = \{x : \langle x, y_j - y_i \rangle \leq \Phi_j - \Phi_i \text{ for all } j\}$$

has nonempty interior. The source paper computes $\nabla^2 E_\nu(\Phi)$ for $\Phi \in U$.

Proof Intuition

The Hessian of E_ν cannot be invertible, because adding $a + \langle b, y_i \rangle$ to the entries Φ_i only translates and rescales the density $e^{-\Phi^*}$; this gives exactly the $(d+1)$ affine null directions. The source regularization adds rank-one terms in precisely those directions. The only possible obstruction would be an extra hidden null direction of the Hessian. The key point is that such a direction would make the normalized densities associated to $\Phi + th$ and $\Phi - th$ agree up to translation to first order. Looking inside the interiors of the Laguerre cells forces h_i to be affine in y_i , so no hidden null direction exists.

Theorem

Theorem 1. *Under the hypotheses above, for every $\Phi \in U$ the matrix*

$$M_\nu(\Phi) = \nabla^2 E_\nu(\Phi) + uu^\top + \sum_{\alpha=1}^d v_\alpha v_\alpha^\top$$

is symmetric positive definite. In particular, the linear system with coefficient matrix $M_\nu(\Phi^{(k)})$ in Algorithm 4.1 of [1] has a unique solution at every iterate $\Phi^{(k)} \in U$.

Proof

Let

$$A = \{(a + \langle b, y_i \rangle)_{i=1}^N : a \in \mathbb{R}, b \in \mathbb{R}^d\} \subset \mathbb{R}^N$$

be the affine-value subspace. Since the support of ν is not contained in a hyperplane, $\dim A = d+1$, and $A = \text{span}\{u, v_1, \dots, v_d\}$.

We first record the kernel statement needed below.

Lemma 1. *For every $\Phi \in U$, the Hessian $H(\Phi) := \nabla^2 E_\nu(\Phi)$ is positive semidefinite and*

$$\ker H(\Phi) = A.$$

Proof. The linear part $\sum_i p_i \Phi_i$ has no second derivative, so $H(\Phi)$ is the Hessian of

$$I(\Phi) = -\log \int_{\mathbb{R}^d} e^{-\Phi^*(x)} dx.$$

The Prekopa–Leindler argument used in Proposition 3.3 of [1] gives convexity of this functional on the finite-dimensional family of lower convex envelopes determined by the values Φ_i . Hence $H(\Phi)$ is positive semidefinite.

For $h \in A$, say $h_i = a + \langle b, y_i \rangle$, one has

$$(\Phi + th)^*(x) = \Phi^*(x - tb) - ta.$$

Thus $I(\Phi + th)$ is affine in t , and therefore $h \in \ker H(\Phi)$. So $A \subseteq \ker H(\Phi)$.

It remains to rule out further null directions. Fix $h \in \mathbb{R}^N$ and, for small t , set

$$\psi_t = (\Phi + th)^*, \quad Z_t = \int_{\mathbb{R}^d} e^{-\psi_t(x)} dx, \quad \rho_t = Z_t^{-1} e^{-\psi_t}.$$

Because $\Phi \in U$, all cells have interior points with strict inequalities; after decreasing $|t|$ if necessary, each support point remains active. Let $\varphi_t = (\Phi + th)^{**}$ be the lower convex envelope of the values $\Phi_i + th_i$ on $\{y_i\}$.

Consider the symmetric second difference

$$D(t) = \frac{E_\nu(\Phi + th) + E_\nu(\Phi - th)}{2} - E_\nu(\Phi).$$

The linear part of E_ν cancels. Moreover, the convex function $(\varphi_t + \varphi_{-t})/2$ is bounded above by φ_0 because it takes values at most Φ_i at all y_i ; therefore

$$\left(\frac{\varphi_t + \varphi_{-t}}{2} \right)^* \geq \varphi_0^*, \quad \mathcal{I} \left(\frac{\varphi_t + \varphi_{-t}}{2} \right) \geq \mathcal{I}(\varphi_0).$$

Applying the quantitative midpoint convexity estimate of [1, Proposition 3.3] to φ_t and φ_{-t} gives, for some translation $z_t \in \mathbb{R}^d$,

$$D(t) \geq c_{\Phi, h, d} \|\rho_t - \rho_{-t}(\cdot - z_t)\|_{L^1(\mathbb{R}^d)}^2$$

for all sufficiently small $|t|$; the value of the positive constant is immaterial.

Suppose now that $h^\top H(\Phi)h = 0$. Then $D(t) = o(t^2)$. Hence, along a sequence $t_n \rightarrow 0$, there are translations z_n with

$$\|\rho_{t_n} - \rho_{-t_n}(\cdot - z_n)\|_{L^1} = o(|t_n|).$$

The density ρ_0 is not translation-invariant in any nonzero direction. More quantitatively, since ρ_0 is a polyhedral log-concave density with finite weighted facet perimeter, the map $r \mapsto \rho_0(\cdot - r\theta)$ has a nonzero L^1 first variation for every $\theta \in S^{d-1}$; compactness of S^{d-1} gives $c > 0$ such that

$$\|\rho_0 - \rho_0(\cdot - r\theta)\|_{L^1} \geq cr + o(r)$$

uniformly in θ . Since $\rho_{\pm t} = \rho_0 + O(t)$ in L^1 , the preceding $o(|t_n|)$ estimate forces $z_n = O(t_n)$. Pass to a subsequence with $z_n/t_n \rightarrow w$.

Let C_i be a compact subset of the interior of $\text{Lag}_i(\Phi)$ with positive Lebesgue measure. On C_i , for n large, the active index in the maximum defining $\psi_{\pm t_n}$ is still i , and also $x - z_n \in \text{Lag}_i(\Phi)$ after shrinking C_i slightly. Writing

$$\bar{h} = \sum_i q_i h_i, \quad q_i = \int_{\text{Lag}_i(\Phi)} \rho_0(x) dx,$$

we have the first-order expansions on C_i :

$$\rho_{t_n}(x) = \rho_0(x) (1 + t_n(h_i - \bar{h}) + o(t_n)),$$

and

$$\rho_{-t_n}(x - z_n) = \rho_0(x) (1 + \langle z_n, y_i \rangle - t_n h_i + t_n \bar{h} + o(t_n)).$$

The L^1 difference being $o(|t_n|)$ forces the coefficient of t_n in the difference to vanish on every cell:

$$2(h_i - \bar{h}) - \langle w, y_i \rangle = 0, \quad 1 \leq i \leq N.$$

Thus $h_i = \bar{h} + \langle w/2, y_i \rangle$, so $h \in A$. This proves $\ker H(\Phi) \subseteq A$, and the lemma follows. $\square$

We now finish the proof of the theorem. For any $h \in \mathbb{R}^N$,

$$h^\top M_\nu(\Phi) h = h^\top H(\Phi) h + (h \cdot u)^2 + \sum_{\alpha=1}^d (h \cdot v_\alpha)^2.$$

By Lemma 1, the first term is nonnegative. If the displayed expression is zero, then $h \in \ker H(\Phi) = A$, while also h is orthogonal to every vector spanning A . Hence $h \in A \cap A^\perp = \{0\}$. Therefore $M_\nu(\Phi)$ is positive definite.

Verification Notes

The proof is analytic. A small independent script, `code/check_d1_newton_matrix.py`, checks the one-dimensional explicit Hessian formula for random full-cell configurations. Run from the repository root:

```
conda run --no-capture-output -n sandbox python \
  runs/fa_banach_001/solutions/partial/2604.09914_newton_matrix_invertibility/\
  code/check_d1_newton_matrix.py
```

On July 18, 2026, it sampled 1500 configurations, found no failures, and reported minimum regularized eigenvalue approximately 0.12924. This numerical check is not used in the proof.

Limitations and Novelty Check

This packet proves only the invertibility of $M_\nu(\Phi)$ for $\Phi \in U$. It does not prove convergence of Algorithm 4.1, does not give lower bounds uniform near the boundary of U , and does not justify any adaptive damping rule. The target is finite-dimensional convex/numerical analysis rather than a core Banach-space structure problem; it was nevertheless retained because the extracted question is precise and the proof is packet-sized.

The bounded local novelty check searched the run registry, solution index, attempts, proof gaps, and ledger for arXiv:2604.09914 and the terms “moment measure”, “Newton matrix”, “damped Newton”, and “Laguerre”. No duplicate packet for this question was found. No broad external web or MathSciNet-style literature search was performed, so novelty confidence is local-to-run rather than global.

Human Review Recommendation

Review the first-order rigidity step in Lemma 1: specifically, the passage from L^1 closeness of ρ_t and a translated ρ_{-t} at order $o(t)$ to the affine formula $h_i = \bar{h} + \langle w/2, y_i \rangle$. This is the only delicate point; the final positive-definiteness argument is then immediate.

References

- [1] G. Bonnet and Y. A. Rubinstein, *Quantitative Stability and Numerical Resolution of the Moment Measure Problem*, arXiv:2604.09914, 2026.